Anatomy of a Futonic Mission (from the HEAD down)

Preprint skeleton — DOCUMENT phase of E-mission-head. Companion to "A First Proof Sprint" (futon6). Format: running example, not monograph — the primary source is `futon6/holes/missions/E-mission-head.md` and its sibling artifacts; this paper reads them rather than repeating them.

Status: skeleton (2026-06-10, Joe + Fable). The §5 synthesis is the novel content; everything else narrates artifacts that already exist.

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1. Two ascents of one mountain

On 2026-06-10 a mission was developed from a single greenfield HEAD to a verified design in one working session — with two instruments watching it the whole way. The first instrument read its **structure**: a live panel showing the mission's typed scopes — phases, sections, patterns, selection records — materializing as each section was written, absent phases rendered as ghosts. The second read its **life**: the same HEAD compiled to an 8+32-bit sigil, birth vitals, falsifiability conditions, a survived adversarial bout, and a deadline clause with teeth.

The Daumal-affine observation: these are two ascents of the same mountain by different faces, and each can report the other's summit. Neither reading reduces to the other; the paper's contribution (§5) is their synthesis.

2. The running example

`E-mission-head.md` — a mission whose subject is missions: "make the mission HEAD a typed object." Its development is unusually well-instrumented because it instrumented *itself*: every phase it grew was captured by the tooling it was building. Primary artifacts (the paper's data):

artifact	reading
`E-mission-head.md` (the doc, HEAD→DOCUMENT)	both — the common sourc
substrate-2 mission-scope hyperedges (31 scopes) + `*mission-overview*`	scope
`E-mission-head.aif.edn` (sigil, vitals, F-nodes, amendments) + `-contra`	AIF
`meme.db` `diagram/E-mission-head` (4+3 arrows) + `cascade/E-mission-head-argue` (3+3)	both
`mission_head_invariants.clj` (VERIFY V0, 5/5)	AIF (verified) / scope (a s
`closure-folds.edn` entries incl. the first `:success false`	the loop's grain

♡ vitals (computed from this document)

confidence 0.33 · xenotype 89% · exotype 00011100

⊗IF ⊗HOWEVER ✓THEN ✓BECAUSE

alive but wobbly: xenotype completeness 0.89; strong anchors ['THEN', 'BECAUSE'], weak anchors ['IF', 'HOWEVER'].

generated 2026-06-10T18:16:29 by `head_exotype_probe.py`

3. The scope reading (anatomy)

The mission as a nested system of typed regions anchored in text: the eightfold spine, loose-sections, pattern/PSR/PUR records, the plain-argument sub-scope; detected (`mission_scope_detect.py`), ingested to substrate-2, rendered as the Scratch-Map. Third-person, structural, *what the mission is made of*. Key device: the **ghost line** — a canon phase with no scope yet. The primary source's opening theme, as the scope typography renders it:

HEAD (Joe, 2026-06-10, verbatim sense)

A mission's HEAD is doing two things at once.
First, in a Jazz or Free Jazz sense, the HEAD is the **theme around which the whole mission is improvised** — ideally in a fully automated way, by following the **patterns of improvisation** that Joe intends to learn from the futon5 work. We are not there yet.
Second, each mission is a **virtual peripheral**, and the HEAD is an **AIF head**: it should be** mappable to the usual AIF terminal vocabulary** — priors, preferences, observations, policies — which would allow us to understand, in typed terms, **what it means to satisfy the requirements of the mission**.
Concretely, the HEAD should therefore carry an extraction of **concepts** — per the **Interest Network** — named entities like "storage", "agents", "patterns", "meme graph", "futon3b", "pattern cascades", obtainable just by *reading the HEAD*. The current kernel extraction is unigram-grade; the upgrade is Interest-Network-grade extraction over the mission corpus, feeding the mission-mode concept lane that already exists. The satisfaction-conditions half connects to E-ground-G: a HEAD mapped to AIF terminal vocabulary is what grounded-G ultimately discharges against.

4. The AIF reading (physiology)

The mission as a lifeform seeded from its HEAD: the recast to pattern form (Golemization), the compiled sigil, health-at-birth, falsifiers authored before constructions, the Calculemus self-argument, the wire-or-die clause. First-person (the mission's own model of itself), dynamical, *what the mission wants and whether it is well*. Key device: the **open arrow** — a typed gap with its RHS specified and its construction owed. The moment the lifeform first spoke — the sigil compilation, from the source:

3.3.1 v0.1 RESULT — the Golemization run (2026-06-10)

The probe's pattern-mode ran on §3.3's recast (same projector, section-wise):

section	bits	conf	nearest anchor	cos
IF	`00011100`	0.28	hexagram-49 **Ge** (molting/revolution)	0.284
HOWEVER	`00011001`	0.36	hexagram-13 **Tongren** (fellowship)	0.306
THEN	`10101100`	0.32	iiching/exotype-248	**0.454**
BECAUSE	`11001000`	0.37	hexagram-31 **Xian** (mutual influence)	**0.461**

xenotype-32 `00011100·00011001·10101100·11001000`, mean-conf 0.33 (baseline 0.29).
Findings, honestly weighted:

1. **Qualitative unlock confirmed: the sigil now exists.** The xenotype was *uncomputable* before the recast; the Golem has a complete shem.
2. **Bit-confidence lift is modest** (+0.04). Sectioning helps; it does not transform the projection. (Caveat: v0.1 reused the whole-pattern projector per section; the bridge's official path trains per-section projectors — some headroom there.)
3. **The real signal moved into anchor proximity, asymmetrically**: THEN and BECAUSE land at cos ≈ 0.46 — a third stronger than the whole-text best (0.349) — while IF/HOWEVER stay weak (≈ 0.29). Reading: **the actionable half of a HEAD (move + warrant) projects into pattern-space well; the intent/gap half is mission-specific prose the pattern-grain anchors don't cover.** Supports next-car 3: HEAD-grain anchors (missions with known outcomes) for the IF/HOWEVER half.
4. The oracle stayed coherent: intent→*molting*, gap→*fellowship*, warrant→*mutual influence*. Recorded as flavor, not evidence.

5. The synthesis (the new thing)

The two readings are not analogues; they are **coupled as observation and model**:

> ***The scope reading is the lifeform's observation space; the AIF reading is its generative model. A mission is well-formed exactly when the two cohere, and the coherence is computable.***

The mapping is exact, not poetic:

scope object	AIF object	the coupling
HEAD section	generative seed (priors)	the compiler (bridge)
eightfold spine	policy stages	the canon is a *prior over trajectories*
ghost line	**prediction error**	canon prior expects a phase; observation (the doc) lacks it
in-passing closure	a satisfied prediction by another path	posterior update without the expected observation
open arrow	expected free energy not yet discharged	A2 prices its persistence
PSR / PUR	policy selection / outcome observation	the pur binder makes the loop legible
falsifier (F-node)	precision channel	what would *count* as error
vitals (health.json)	interoception	computed FROM the scope-readable doc
the panel	exteroception	the body as seen

Three consequences worth the paper:

1. **The body-schema claim.** The H1–H3 wiring (health computed from the doc → displayed in the panel → seeded into the head) is not plumbing; it is the *binding* of interoception to exteroception — the point at which the mission acquires a body schema. Before it, two instruments; after it, one organism with two senses.
2. **Coherence is checkable, so DOCUMENT can be earned.** Disagreements between the readings are findings, mechanically surfaced: a scope the model doesn't want (decorative structure), a want with no scope (unobservable desire — exactly the unminted-pattern catch), a

ghost the model says is satisfied (the in-passing DOCUMENT closure). The session produced instances of all three.

3. **The synthesis discharges the cascade's own largest gap.** The mission's ARGUE cascade carries the unminted candidate `two-projections-of-one-quantity`. The synthesis *is* that pattern's construction: scope-reading and AIF-reading as two projections of one underlying typed-arrow object (substrate-2 hyperedges + meme.db arrows + the package are three stores of *one* structure). Publishing the paper mints the pattern; the preprint is its flexiarg's BECAUSE.

And the Daumal closure, which the stack supplies on its own: the points where two independent ascents *agree* are the only points one may call real — and the stack already has a name and a discipline for externally-checkable value found on the mountain: **the peradam**. The synthesis predicts where peradams form: at reading-agreements. "One descends, one sees no longer, but one has seen."

And the argument in the register anyone can read — the mission's own plain-language statement, boxed as it appears in the source:

4.1 Plain-language argument (the version anyone can read)

When we start a piece of work, we write a short statement of what it is for. Right now that statement is just words: nothing else in the system can read it, check it, or use it. So nobody can tell, at the start, whether a piece of work rests on solid ground or shaky ground — and when work quietly stalls, nothing notices. This has already happened here: a major piece of work was "finished" on paper three months ago, and nobody saw that it was never actually connected to anything.
The proposal: when work begins, turn its opening statement into a small summary the system can read — what the work wants, what is in the way, what it will do, and why. From that, the system can show, from day one, how strong or weak the work looks, what is missing, and what would count as finishing. As the work proceeds, the same readout shows whether it is getting healthier.
What we checked before believing this: we tried it on this very piece of work. It partly worked — the "what to do" and "why" parts came through clearly; the "what we want" part needs better reference material, which we now know how to gather. We also put up the strongest objections we could find, and answered each one:

1. *"These summaries will be built and then ignored, like last time."* Last time, nothing showed that the finished work was never put to use. This time there is a deadline: if the pieces are not in use by then, the work is marked as failed, in a place everyone can see.
2. *"People will write their statements to score well."* The readout is never used as a target or a gate. It is information only, checked occasionally and independently.
3. *"This will make creative work bureaucratic."* The summary describes only the starting point, never the path. The path stays free — that freedom is the point of working this way, and it is kept on purpose.

Every claim above comes with a stated test that would prove it wrong, and the first of those tests has already been run and passed.
Why this is needed, in one line: **work should be able to say what it is for, in a form that lets anyone — or anything — check how it is doing.**

6. Method appendix (planned)

The session as the Cyborg learning loop run at operator scale: fold records (incl. the first `:success false`), the closure standard, the bell-handoff INSTANTIATE, the review gates. Cross-refs: E-scope-audit (the instrument's own audit trail), C-falsifiable-missions §7 (the closure schema), E-ground-G (what the satisfaction conditions discharge against).

7. Relation to "A First Proof Sprint"

The Proof Sprint documents a *proof* finding its way through the stack; this paper documents a *mission* doing the same — one level up: the unit of work examining the unit-of-work concept. Same house, different floor.

TODO (next sessions): §3/§4 prose passes over the primary source; figures (the panel at frames 1→6; the bout's cross-edge diagram; the vitals card); §5 worked dis/agreement instances with store queries; decide venue + whether the two-projections flexiarg ships with the paper or before it.